

Retail Discount Profitability & Revenue Leakage Audit

Is discounting functioning as a strategic promotional tool, or has it become a structurally embedded pricing policy that continuously compresses margins?

ANALYST

Zeeshan Akram

STACK

PostgreSQL · Python (Pandas) · Power BI

DATASET

Contoso Retail (Synthetic)

Executive Verdict

Five headline findings from the full-stack audit (SQL · Python · Power BI)

\$218.8M

Total Revenue

Business is profitable in aggregate

61%

Discount Penetration

Of all ~223K order lines

54%

Discounted Margin

vs 58% at full price

\$97

Profit Gap Per Line

Discounted vs full-price lines

4–5pp

Margin Compression

Across low-to-high discount tiers

01

Discounting is the majority pricing behavior. 61% of all order lines are sold below listed price. It is not a promotional exception.

02

Margin compresses consistently as discount depth increases, from 54% at the low tier to 49% within the 10–20% tier, with no recovery at any level.

03

Computers and Cell Phones account for ~46% of all high-discount order lines. Their scale amplifies a system-wide pattern into the majority of total margin erosion.

04

High-discount customers produce the lowest margins (50.4%) and the lowest purchase frequency (3.2 avg. order lines). They are the least commercially efficient segment.

05

Discount behavior is uniform across all stores and has been stable year over year. This is a centrally governed structural policy, not a local or temporary pattern.

AUDIT VERDICT: Discounting has ceased to function as a tactical lever. It operates as the company's structural pricing floor.

Transaction-Level Findings

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Discount depth drives consistent, directional margin compression across all tiers

61%

Discount Penetration

Of all ~223K order lines

4pp

Margin Efficiency Gap

Discounted vs full-price

\$97

Profit Per Line Gap

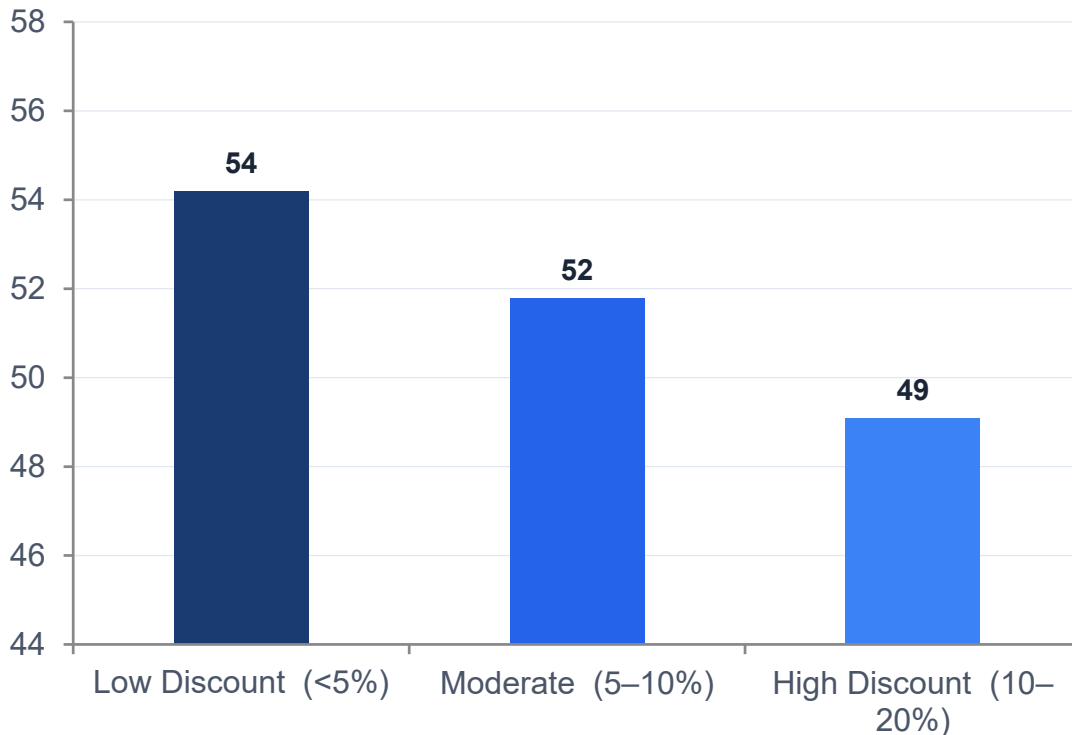
\$508 vs \$605 per order line

-0.37

Discount-Margin Correlation

Moderate, systematic, directional

Average Margin % by Discount Tier

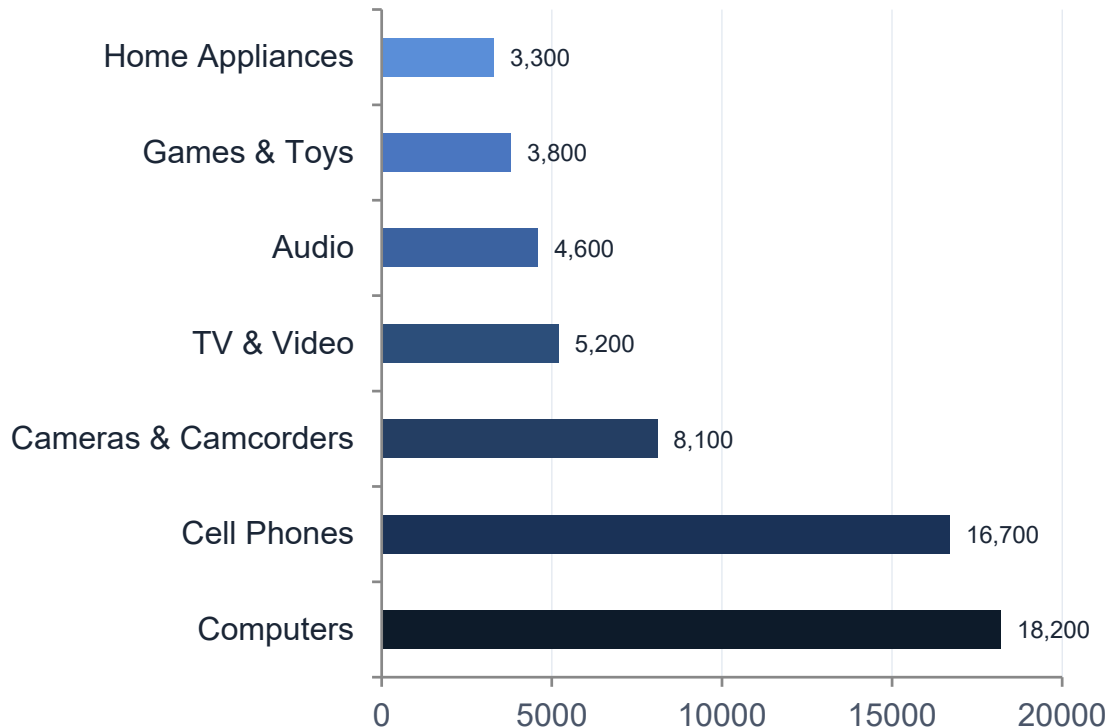


Category-Level Leakage

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Volume amplification, not discount severity, drives the largest aggregate margin impact

High-Discount Order Lines by Category (10–20% tier)



~46%

of all high-discount volume

Computers and Cell Phones combined. These two categories alone concentrate nearly half of all high-discount transaction activity.

4.4–5.2pp

margin compression per category

Consistent across every product category when moving from low to high discount levels. The compression rate is uniform, no category is exempt.

KEY IMPLICATION

A pricing decision in Computers or Cell Phones carries outsized impact relative to any other product group. This concentration is the primary point of intervention.

Customer Behavior Findings

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Discounting has become embedded customer expectation, not a targeted incentive

LOW DEPENDENCY

< 50% of purchases discounted

21,336

Customers

53.5%

Avg Margin

31%

Avg Discount Rate

3.77

Avg Order Lines

MEDIUM DEPENDENCY

50% – 75% of purchases discounted

14,758

Customers

51.7%

Avg Margin

66%

Avg Discount Rate

6.23

Avg Order Lines, highest of all segments

HIGH DEPENDENCY

> 75% of purchases discounted

16,095

Customers

50.4%

Avg Margin

96%

Avg Discount Rate

3.20

Avg Order Lines, lowest of all segments

The Engagement Finding (Inverted-U Pattern)

Medium-dependency customers generate 6.23 avg. order lines, nearly double the High-dependency group (3.20). Beyond a threshold, additional discount dependency reduces engagement rather than increasing it. High-discount customers are commercially inefficient in two dimensions simultaneously: lower margins and lower purchase frequency.

24.2% of Customers at Near-Total Discount Dependency

24.2% of customers transact at 90%+ discount usage. Only 11% purchase near full price. For this segment, the discount is no longer an incentive, it is the effective price. Full-price purchasing has ceased to be a default expectation for a material share of the customer base.

Strategic Recommendations

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Operational response: ranked by priority, expected impact, and implementation complexity

Recommendation	Priority	Expected Impact	Implementation
Define and enforce a maximum discount depth cap, contain activity within the 10–20% tier	High	Margin recovery of 4–5pp on affected lines	Low: policy change
Apply category-specific discount rules to Computers and Cell Phones	High	Reduces ~46% of high-discount volume immediately	Medium: pricing update
Introduce segment-based promotional eligibility across Low / Medium / High customer tiers	Medium	Reduces High-segment dependency over time	Medium: CRM integration
Set the 61% discount rate as an actively managed KPI with a defined acceptable range	Medium	Converts passive outcome into a controlled variable	Low: reporting change
Conduct a full central pricing policy review	Medium	Any change propagates across all stores at full scale	High: policy redesign

STRATEGIC NOTES

Caution with High-dependency customers: Any rapid move toward full-price enforcement for this segment carries churn risk. A phased approach is advisable.

Store interventions will have no effect: Pricing is centrally governed. Any change must occur at the policy level to propagate at full scale.

No urgent deterioration: Discount rates and margins are stable year over year. The business is not in a declining trend, but it is operating at a structurally compressed baseline.

CONCLUSION

The Audit in One Statement

The business is not in financial distress. At the aggregate level, it is profitable. What this audit finds is that the current margin level is not a temporarily suppressed version of a higher baseline, it is the baseline, built into the pricing model through discounting behavior that has become self-reinforcing over time. Reversing this requires deliberate policy decisions, not operational fixes.

01

The pricing floor has shifted.

The 52% blended margin is the operating reality under current pricing policy. Absent deliberate change, it will remain there.

02

Volume concentration creates asymmetric exposure.

Computers and Cell Phones carry outsized risk. Targeted policy changes in these two categories would address the majority of total margin erosion.

03

The solution space is at the policy level.

Uniform store behavior and stable year-over-year trends confirm this is a centrally governed structure. Any meaningful change must begin at that level.